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检测
TESTING
CNAS L4743

检测报告

报告号: AJFS2108007147RS_CN 日期: 2021 年 8 月 27 日 第 1 页 共 12 页

北京天山新材料技术有限公司

北京八大处高科技园区双园路 5 号

此报告是报告号为 AJFS2108007147RS 的中文译本, 如中英文有差异, 概以英文为准。

样品名称: 聚氨酯粘接剂

SGS 参考号: TJRS2021081267

颜色: 黑色

型号/货号: 1956

以上测试样品数据和信息为客户提供并确认, SGS 不对这些数据和信息的真实性、完整性以及由此引发的结果和/或结论的有效性负责。检测结果适用于收到的样品。

测试要求:

EN 45545-2:2013+A1:2015 铁路应用 铁路车辆防火 第 2 部分: 材料及部件的防火性能要求, 测试参照表格 5—材料要求 (R1)。

测试结果: -- 参见数据附表 --

测试持续时间:

样品收到日期 : 2021 年 8 月 20 日

测试执行时间 : 2021 年 8 月 20 日 至 2021 年 8 月 26 日

通标标准技术服务有限公司安吉分公司

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邹石龙

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I. 样品信息

样品描述	铝板+胶+铝板
颜色	银色
暴露（测试）面	任意面
样品尺寸	T02 ISO 5658-2: 约 800mm×155mm×2.6mm
	T03.01 ISO 5660-1: 约 100mm×100mm×2.6mm
	T10.01/T10.02/T11.01: 约 75mm×75mm×2.6mm
基材	铝板 / 银色 / 1.0mm

II. 结果概要

要求简称(适用部分)	参考测试方法	参数单位	测试结果 *
R1 (IN1A; IN1B; IN1D; IN1E; IN4; IN5; IN6A; IN7; IN8; IN9B; IN11; IN12A; IN12B; IN14;F5)	T02 ISO 5658-2	CFE kW/m ²	50.0
	T03.01 ISO 5660-1: 50 kW/m ²	MARHE kW/m ²	4.3
	T10.01 EN ISO 5659-2: 50 kW/m ²	D _s (4) 无量纲	0.3
	T10.02 EN ISO 5659-2: 50 kW/m ²	VOF ₄ min	0.4
	T11.01 EN ISO 5659-2: 50 kW/m ²	CIT _G 无量纲	0.004

* 测试详细信息，参见本报告的附录。

III. 结论

根据测试结果，送检样品符合R1（详见EN 45545-2:2013+A1:2015表格5）中有关HL3危险等级。

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测试要求 EN 45545-2:2013+A1:2015 表 5 材料综合要求 (R1)

要求简称 (适用部分)	参考测试方法	参数单位	定义要求	HL1	HL2	HL3
R1 (IN1A; IN1B; IN1D; IN1E; IN4; IN5; IN6A; IN7; IN8; IN9B; IN11; IN12A; IN12B; IN14; F5)	T02 ISO 5658-2	CFE kW/m ²	最小	20 a	20 a	20 a
	T03.01 ISO 5660-1: 50 kW/m ²	MARHE kW/m ²	最大	a -	90	60
	T10.01 EN ISO 5659-2: 50 kW/m ²	D _s (4) 无量纲	最大	600	300	150
	T10.02 EN ISO 5659-2: 50 kW/m ²	VOF ₄ min	最大	1200	600	300
	T11.01 EN ISO 5659-2: 50 kW/m ²	CIT _G 无量纲	最大	1.2	0.9	0.75

a 根据EN 45545-2:2013+A1:2015条款5.3.7, 进行ISO 5658-2测试时, 如果出现燃烧的滴落物或碎片, 或者在特殊情况下ISO 5658-2测试材料未点燃, 而结果为不能分级, 应增加如下要求:

根据EN ISO 11925-2, 施加30s点火时间。

可接受的条件是

--- 60s内火焰传播 < 150mm;

--- 没有燃烧的滴落物或者碎片

陈述:

此符合性声明仅基于本次实验室活动的实际值, 未将本次实验室活动的结果不确定度影响计入。

本试验结果得出的产品性能是在特殊条件下的检验结果。在实际应用中, 它们不能单独作为评价该产品潜在火灾危险性的依据。

本测试结果仅代表被测试试样。

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附录 1: T02 ISO 5658-2:2006 侧面火蔓延试验

1. 预处理

T: 23±2°C, R.H: 50±5%条件下, 样品处理至重量恒定状态。

2. 测试结果

样品编号:	1		持续燃烧热值 (MJ/m²)	2		持续燃烧热值 (MJ/m²)	3		持续燃烧热值 (MJ/m²)
重量 (g)	683.3			690.2			699.7		
点燃时间: (min:sec)	未点燃			未点燃			未点燃		
传播的时间	min	sec		min	sec		min	sec	
50 mm	-	-	-	-	-	-	-	-	
100 mm	-	-	-	-	-	-	-	-	
150 mm	-	-	-	-	-	-	-	-	
200 mm	-	-	-	-	-	-	-	-	
250 mm	-	-	-	-	-	-	-	-	
300 mm	-	-	-	-	-	-	-	-	
350 mm	-	-	-	-	-	-	-	-	
400 mm	-	-	-	-	-	-	-	-	
450 mm	-	-	-	-	-	-	-	-	
500mm	-	-	-	-	-	-	-	-	
550 mm	-	-	-	-	-	-	-	-	
600 mm	-	-	-	-	-	-	-	-	
650 mm	-	-	-	-	-	-	-	-	
700 mm	-	-	-	-	-	-	-	-	
750 mm	-	-	-	-	-	-	-	-	
测试时间 (min:sec)	10:00		10:00		10:00		10:00		
最终传播距离 (mm)	-		-		-		-		
CFE (kW/m²)	50.0		50.0		50.0		50.0		

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样品编号:	4		持续燃烧热值 (MJ/m ²)	5		持续燃烧热值 (MJ/m ²)	6		持续燃烧热值 (MJ/m ²)
重量 (g)	689.0			/			/		
点燃时间: (min:sec)	未点燃			/			/		
传播的时间	min	sec		min	sec		min	sec	
50 mm	-	-	-	/	/	/	/	/	/
100 mm	-	-	-	/	/	/	/	/	/
150 mm	-	-	-	/	/	/	/	/	/
200 mm	-	-	-	/	/	/	/	/	/
250 mm	-	-	-	/	/	/	/	/	/
300 mm	-	-	-	/	/	/	/	/	/
350 mm	-	-	-	/	/	/	/	/	/
400 mm	-	-	-	/	/	/	/	/	/
450 mm	-	-	-	/	/	/	/	/	/
500mm	-	-	-	/	/	/	/	/	/
550 mm	-	-	-	/	/	/	/	/	/
600 mm	-	-	-	/	/	/	/	/	/
650 mm	-	-	-	/	/	/	/	/	/
700 mm	-	-	-	/	/	/	/	/	/
750 mm	-	-	-	/	/	/	/	/	/
测试时间 (min:sec)	10:00		/		/		/		
最终传播距离 (mm)	-		/		/		/		
CFE (kW/m ²)	50.0		/		/		/		

测试观察: 无

根据每一个试样的试验数据计算出下表参数

参数	样品编号				均值
	1	2	3	4	
平均持续燃烧热值(Qsb) (MJ/m ²)	-	-	-	-	-
火焰熄灭时的临界辐射通量 (CFE) (kW/m ²)	50.0	50.0	50.0	50.0	50.0

注: “-”表示火焰未传播至 150mm 处。

“/”表示未进行测试。

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附录 2: T03.01 ISO 5660-1:2015 对火的反应试验 热释放、烟量和质量损失速率 第 1 部分: 热释放速率 (锥形量热法) 和产烟速率 (动态测量), 辐射照度 50 kW/m²

1. 预处理

T: 23±2°C, R.H: 50±5%条件下, 样品处理至重量恒定状态。

2. 测试结果

测试方向	水平			
样品暴露面积 / m ²	0.0088			
辐射照度 / (kW/m ²)	50			
加热锥底板与试样上表面之间的距离 / (mm)	25			
样品编号	1	2	3	平均值
样品初始质量 / g	53.6	53.9	54.1	53.9
点燃时样品质量 / g	53.0	52.5	53.2	52.9
剩余样品质量 / g	52.2	51.9	52.4	52.1
样品质量损失 / g/m ²	160.0	220.0	190.0	190.0
平均质量损失速率 $\dot{m} / (g \cdot m^{-2} \cdot s^{-1})$	0.2	0.4	0.3	0.3
单位面积平均质量损失速率 $\dot{m}_{A,10-90} / (g \cdot m^{-2} \cdot s^{-1})$	0.3	0.3	0.2	0.3
持续火焰发生时间 / s	531.0	613.0	637.0	593.7
是否重新插入点火器 ¹⁾	否	否	否	---
热释放速率峰值 / (kW/m ²)	34.9	27.1	20.7	27.5
引燃后单位面积 180s 平均热释放速率 / (kW/m ²) ²⁾	16.0	16.2	14.3	15.5
引燃后单位面积 300s 平均热释放速率 / (kW/m ²)	11.2	10.1	11.0	10.7
热释放总量 / (MJ/m ²)	3.9	3.6	4.2	3.9
平均有效热释放 / (MJ/kg)	43.1	52.8	42.6	46.2
引燃前, 单位面积总产烟量 / m ² m ⁻²	0.4	1.1	3.2	1.6
引燃后, 单位面积总产烟量 / m ² m ⁻²	9.2	7.2	13	9.8
测试持续时间 / s ³⁾	1200.0	1200.0	1200.0	1200.0
最大平均热释放速率 (MARHE) / KW/m ²	4.7	4.1	4.1	4.3
测试现象 ⁴⁾	--			
特殊的安装程序 ⁵⁾	f			

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备注:

- 1) 如果在移开火花器60s内, 火焰熄灭, 在5s内再次插入火花器, 并保持到测试结束。
- 2) 应对3个试样在180s内的平均热释放速率进行比较, 若其中的一个与这3个的平均值之差超过10%, 则应另取3个试样进行试验, 除非其平均值小于10 kW/m²。
- 3) 试验终止条件:
 - ☐ a. 持续燃烧32分钟 (30分钟试验时间加2min试验后追加数据搜集时间);
 - ☐ b. 样品30分钟内未点燃;
 - ☐ c. XO₂ 回到试验前氧气浓度值100 μl/l, 持续10分钟。10分钟开始时刻为试验结束时间;
 - ☐ d. 持续60秒, 样品质量少于0.1g, 60秒开始的时间为试验结束时间;
 - ☒ e. 20分钟 (EN 45545-2:2013+A1:2015, T03.01 ISO 5660-1中所述)。
- 4) 观察和记录其他测试现象, 如熔化、膨胀、变形爆裂等。
- 5) 采用的特殊安装程序:
 - ☐ a. 如果试样因膨胀或变形, 在引燃前接触到火花塞, 或引燃之后接触到辐射锥下表面, 这种情况下应使辐射锥的下表面和试样表面有60mm的间隔;
 - ☐ b. 其他尺寸不稳定的制品 (例如试验期间卷曲或收缩), 应采用4根金属丝限制其过分的变形。用一根金属丝将试样安装架和定位架组件缠紧, 并保证与组件的4个边之一平行, 且距离约为20mm.将金属丝两端拧紧, 使金属丝和定位架固定。试验前去掉金属丝多余部分。其余3根金属丝应以相同的方式固定, 并分别与其余3条边平行;
 - ☐ c. 如果4根铁丝无法控制样品的剧烈变形, 测试时需要使用直径 (0,8 ± 0,1) mm, 间距 (20 ± 2) mm的铁丝网;
 - ☐ d. 样品膨胀熔化成液态会导致熔融物溢出边框, 或者填充到样品保持架和样品架之间, 这种测试结果是无效的。因此这类材料测试时不能使用样品保持架, 并封装在0.1mm厚的铝箔托盘内 (边缘超出样品顶部10mm);
 - ☐ e. 材料如纤维, 需要物理约束或者压缩至一定的安装密度, 测试时需使用直径(1,0 ± 0,1) mm, 间距 (9 ± 1) mm构成的铁丝支架结构
 - ☒ f. 没有使用特殊的安装程序
- 6) 图1: 单位面积热释放速率曲线(kW/m²)。
- 7) 孔板系数为 0.043; 排气流量为 0.024 ± 0.002 m³/s。

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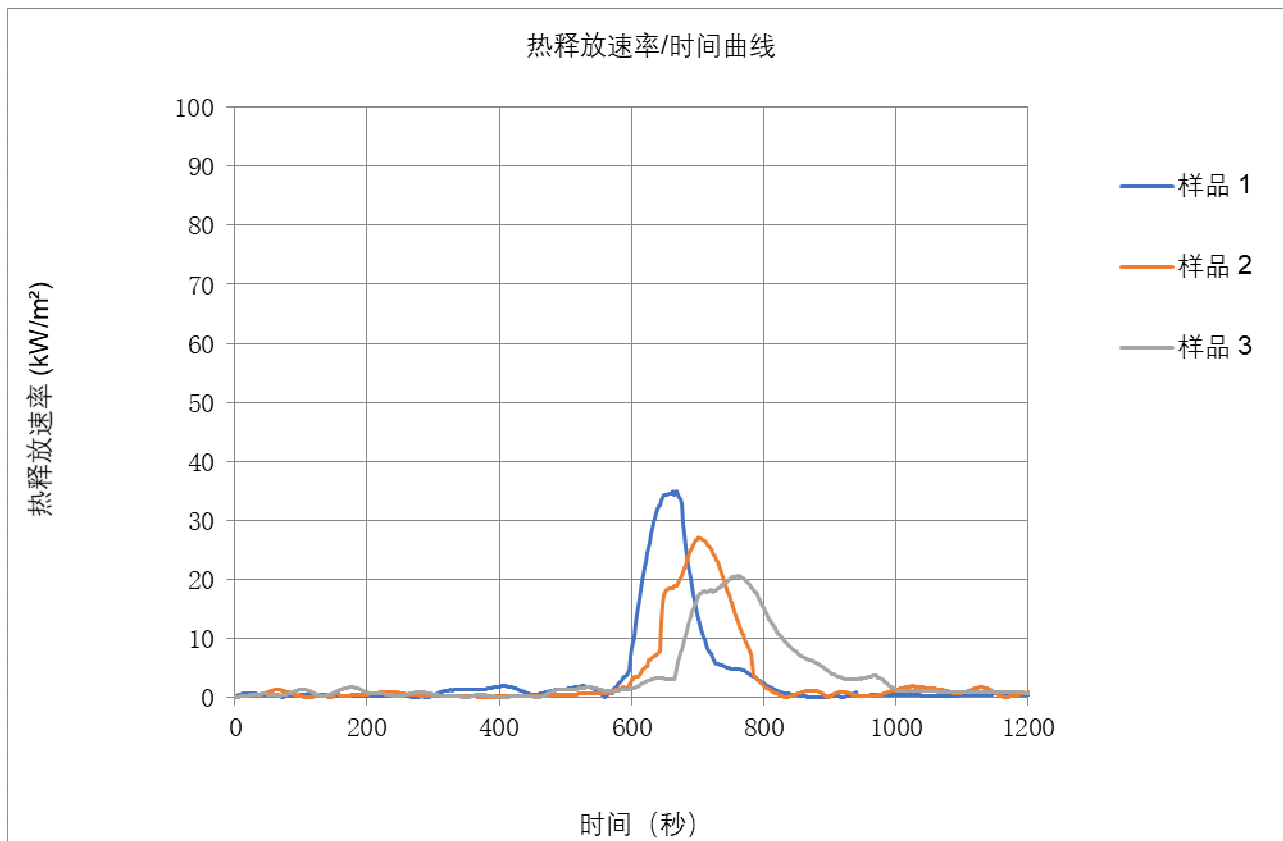


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检测报告

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曲线图:



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附录 3: T10.01&T10.02 EN ISO 5659-2:2017 塑料 烟生成 第 2 部分:单室法测定烟密度试验方法, 辐射照度 50kW/m² 无引燃火焰模式, 测试时间 10min.

1. 预处理

T: 23±2°C, R.H: 50±5%条件下, 样品处理至重量恒定状态。

2. 测试结果

参数	1	2	3	平均值
重量 (g)	30.0	30.3	30.4	30.2
D _s (1.5)	0.1	0.2	0.1	0.1
D _s (4)	0.3	0.3	0.2	0.3
D _s (10)	1.0	1.3	0.8	1.0
VOF ₄ min	0.3	0.6	0.4	0.4
D _s max	1.0	1.3	0.8	1.0
T (D _s max) s	600	598	600	/

注释:

D_s (n) — 试验进行第n分钟时的比光密度;

VOF₄ — 试验前4分钟的比光密度累加值;

D_s max — 试验期间最大比光密度。

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附录 4: T11.01 EN 45545-2:2013+A1:2015 附录 C 采用 FTIR 技术, 进行 EN ISO 5659-2 烟箱法气体分析
辐射照度 50kW/m² 无引燃火焰模式, 测试时间 10min.

1. 预处理

T: 23±2°C, R.H: 50±5 % 条件下, 样品处理至重量恒定状态。

2. 测试结果

1) 测试开始4分钟

气体	1	2	3	均值
二氧化碳 (CO ₂)	23.0	27.2	29.3	26.5
一氧化碳 (CO)	1.3	1.0	1.0	1.1
氟化氢 (HF)	ND	ND	ND	---
氯化氢 (HCl)	ND	ND	ND	---
溴化氢 (HBr)	ND	ND	ND	---
氰化氢 (HCN)	ND	ND	ND	---
氮氧化物 (NO _x)	1.6	ND	3.2	1.6
二氧化硫 (SO ₂)	ND	ND	ND	---

2) 测试开始8分钟

气体	1	2	3	均值
二氧化碳 (CO ₂)	28.4	40.4	42.7	37.2
一氧化碳 (CO)	3.0	2.6	3.0	2.9
氟化氢 (HF)	ND	ND	ND	---
氯化氢 (HCl)	ND	ND	ND	---
溴化氢 (HBr)	ND	ND	ND	---
氰化氢 (HCN)	ND	ND	ND	---
氮氧化物 (NO _x)	2.6	ND	2.2	1.6
二氧化硫 (SO ₂)	ND	ND	ND	---

注: ND---未发现

所有数值单位为mg/m³。

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3) 计算 CIT_G

气体成分	参考浓度 mg/m^3
二氧化碳 (CO_2)	72 000
一氧化碳 (CO)	1 380
氟化氢 (HF)	25
氯化氢 (HCl)	75
溴化氢 (HBr)	99
氰化氢 (HCN)	55
氮氧化物 (NO_x)	38
二氧化硫 (SO_2)	262

$$CIT_G = 0.0805 \cdot \sum_{i=1}^{i=8} \frac{C_i}{C_i}$$

其中

CIT_G — 常规毒性指数;

C_i — 第 i 类气体浓度;

C_i — 第 i 类气体参考浓度。

参数	1	2	3	均值
CIT_G (4 min)	0.004	0.000	0.007	0.003
CIT_G (8 min)	0.006	0.000	0.005	0.004

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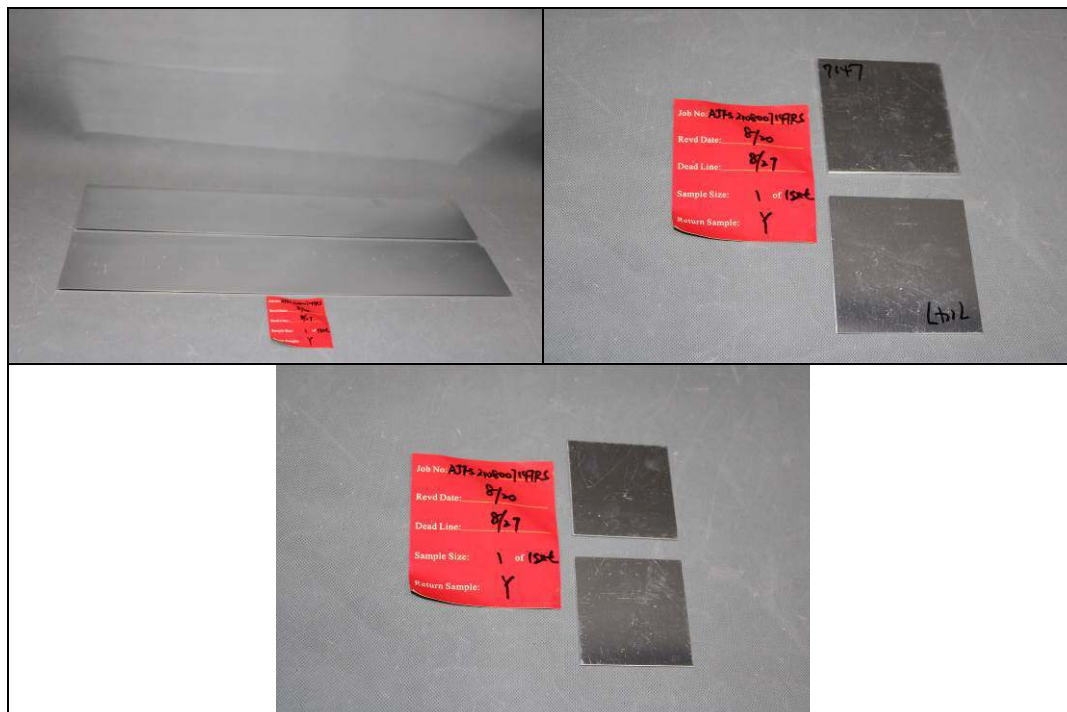
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